C Programming Notes

Topic: Program Building Blocks

1 C Program Structure

Every C program follows a basic structure. Understanding this helps you write programs correctly.

```
#include <stdio.h>    // Preprocessor command

int main() {          // Main function
    // Statements & Expressions
    return 0;         // Return statement
}
```

- Preprocessor Commands: Prepare program before compilation.
- Main Function (main()): Starting point of execution.
- Statements & Expressions: Instructions for the computer.
- Return Statement: Ends program and returns a value to the OS.

2 Preprocessor Commands

Preprocessor commands are special instructions that start with `#` and are executed before the compilation of the program. They help prepare the program for compilation by including libraries, defining constants, or controlling which code gets compiled.

a) `#include`

The `#include` command is used to add external libraries or header files to your program. Libraries contain pre-written code, like functions, that you can use.

- `<stdio.h>` – Standard Input/Output library (for functions like `printf()` and `scanf()`).
- `<math.h>` – Math library (for functions like `sqrt()`, `pow()`, etc.).

```
#include <stdio.h>    // Allows using printf(), scanf(), etc.
#include <math.h>     // Allows using sqrt(), pow(), etc.
```

b) `#define`

The `#define` command is used to create constants or macros. Once defined, these values cannot be changed during program execution.

- Constant example: `#define PI 3.14159` – PI always represents 3.14159.
- Macro example: `#define AREA(r) (PI * r * r)` – AREA(r) calculates the area of a circle for any radius `r`.

```
#define PI 3.14159
#define AREA(r) (PI * r * r)
```

c) `#undef`

The `#undef` command removes a previously defined macro or constant. After this, the macro can no longer be used in the program.

```
#define PI 3.14    // Define PI
#undef PI            // Remove PI definition
```

d) `#ifdef` / `#ifndef`

These commands are used for conditional compilation. They allow certain parts of code to run only if a macro is defined or not defined.

- `#ifdef MACRO` – Code inside executes only if MACRO is defined.
- `#ifndef MACRO` – Code inside executes only if MACRO is not defined.

```
#define TEST

#ifdef TEST
    printf("TEST is defined\\n");   // Executes because TEST is
#endif

#ifndef TEST2
    printf("TEST2 is not defined\\n"); // Executes because TEST2
#endif
```

e) `#pragma`

The `#pragma` command is used to give special instructions to the compiler. These are often advanced commands and may vary depending on the compiler.

- Example: `#pragma once` – Ensures a file is included only once in a program, preventing duplicate definitions.

```
#pragma once   // Ensures the file is included only once
```

3 Functions

A function is a block of code performing a specific task. Functions help organize and reuse code.

```
int add(int a, int b) {
    return a + b;
}

int main() {
    int sum = add(5, 3);
    printf("Sum = %d", sum);
    return 0;
}
```

- Return Type: Type of value returned (e.g., int, float).
- Parameters: Inputs to the function.
- Function Name: Name to call the function.

4 Variables

Variables store data in memory. They are like containers.

```
int age = 15;
float height = 1.65;
char grade = 'A';
```

- int: Whole numbers
- float: Decimal numbers
- char: Single characters

5 Statements & Expressions

Statements are complete instructions; expressions are combinations of variables/operators producing a value.

```c
int a = 5;       // Statement
int b = 10;
int sum = a + b;  // Expression
printf("%d", sum); // Statement
```

6 Comments

Comments are notes for programmers, ignored by the compiler.

Single-line comment

```c
// This is a single-line comment
```

Multi-line comment

```c
/*
This is a multi-line comment
It can span multiple lines
*/
```

💻 Classroom Activities

Activity 1: Identify Program Parts

```c
#include <stdio.h>

int main() {
    printf("C Programming is fun!");
    return 0;
}
```

Label: preprocessor, main function, statements, return statement.

Activity 2: Write Your Own Program

```c
#include <stdio.h>
#define SCHOOL "ABC School"

int main() {
    printf("My name is [Your Name]");
    printf("Welcome to %s", SCHOOL);
```

```
    return 0;
}
```

🏠 Homework / Practice

Write a program that defines `#define COUNTRY "Rwanda"` and prints:

```
Hello from Rwanda!
```